# Technical Proposal

## Cost-Aware Design of Seawater-Resistant Copper Alloy Powders for Blown-Powder Directed Energy Deposition

Office of Naval Research (ONR)
Fiscal Year 2027 Young Investigator Program (YIP)

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**Contents**


**1 Technical Approach** **1**

1.1 Problem and objectives . . . . . 1
1.2 Copper alloys in naval seawater service . . . . . 2
1.3 Scientific concepts . . . . . 4
1.4 Comparison with other approaches . . . . . 5
1.5 Preliminary results . . . . . 6
1.6 Method advances . . . . . 8
1.7 Hypotheses and rejection criteria . . . . . 9
1.8 Research plan . . . . . 10
1.9 Expected results and new knowledge . . . . . 14
1.10 Future Naval Relevance . . . . . 15
1.11 Project Schedule and Milestones . . . . . 16
1.12 Goals and Benchmarks . . . . . 16
1.13 Gold Standard Science . . . . . 16

**2 Management Approach** **16**

**3 Principal Investigator Qualifications** **19**

**Principal Investigator's Curriculum Vitae (excluded from page limit)** **CV-1**

**Letter of Support (excluded from page limit)** **L-1**

**4 Responsibility** **20**

**5 Data Management Plan** **D-1**

**Current and Pending Support (excluded from page limit)** **S-1**

**Cited References (excluded from page limit)** **R-1**

## 1. Technical Approach

> *"Over the last year, we placed nearly 120 additive manufacturing parts into development, installed 20 of those parts, and eliminated more than 1,400 cumulative days of delay."*
>
> Adm. Daryl Caudle, Chief of Naval Operations, 2026 [1]

### 1.1 Problem and objectives

In the Navy, much of the seawater service runs on two kinds of copper alloy. Copper-nickel alloys, 90-10 (UNS C70600) and 70-30 (UNS C71500), carry seawater in piping, condensers, and heat exchangers, and nickel-aluminum bronze (NAB, UNS C95800) is cast into propellers, valves, and pump parts [2, 3, 4]. Both were developed for casting or wrought processing. Directed energy deposition (DED), in which a laser melts a stream of powder blown into a moving melt pool, can build large parts and restore worn or damaged ones without a foundry [5, 6]. DED, however, cools each layer from the melt within seconds and reheats it many times as later layers pass, so it produces microstructures these alloys were never designed for. In NAB, fast cooling can retain a martensitic phase that corrodes selectively in seawater [7, 8], and the Navy specification for NAB deposited by wire-arc DED requires the same 6-hour anneal as castings [9]. That step is costly for a large part and impractical for many repairs made in place. Navy specifications for printed copper alloys all use existing compositions [9, 10, 11], and the design of compositions for laser deposition has only begun [12].

Designing one runs through a chain of steps: a thermodynamic calculation, a powder, deposited test specimens, microscopy, and property tests. The costs of the steps differ greatly. A thermodynamic calculation takes minutes, a powder lot a day, a DED session days including turnaround, and a corrosion immersion weeks. The protective film that gives copper-nickel its resistance takes weeks to months to mature [13, 14]. Experimental campaigns often send each candidate through the whole chain, spending much of the budget on full tests of candidates that a cheap early measurement could have ruled out. Computational campaigns often work as a funnel: many simulations, then a few candidates passed on to experiments. If the simulation misses a physical effect, the funnel discards good candidates before any experiment can catch the error.

The powder is a second obstacle. Single-stream blown-powder DED, the form in use today, feeds a pre-alloyed powder from one hopper. Each new composition therefore needs its own powder lot, in a size range that flows through the nozzle and that the melt pool catches [15, 16]. The lot must also fill the hopper, about 2 kg on the larger of the two systems this project will use (Section 1.8.2). Industrial gas atomizers make powder in lots of tens of kilograms or more, so new compositions are rarely tried [TODO Sterling: cite a minimum lot size, e.g., a vendor's]. Our ultrasonic atomizer makes a lot of a new composition in a day from about half a kilogram of feedstock, and 2 kg in about six runs. Ultrasonic atomizers have already made powder for blown-powder DED from NAB propeller grinding chips [17, 18] and from copper cable scrap [19], but only in existing compositions. That turns the powder from a purchase into steps the search can choose. A small lot is enough for cheap deposits of many compositions on a small DED system near BYU. A full lot is made only for those worth building on the larger system. It also lets the search design the powder along with the alloy.

This project asks a basic question. When the steps of a materials workflow differ in cost by orders of magnitude, and results arrive at different times, how should one choose the next measurement and the next candidate? It also asks a materials question. How do composition and the DED thermal history together set segregation, cracking, fusion to a substrate, and corrosion in an as-deposited copper alloy? We answer both on a concrete design problem: a pre-alloyed copper alloy powder for single-stream blown-powder DED. The powder must deposit without cracks or pores, on itself and on the alloys it would repair, and the deposit must resist seawater without heat treatment at least as well as 70-30 copper-nickel. Repair of cast NAB and wrought copper-nickel is the demanding test of both answers. The project has five objectives:

1. Extend a cost-aware search method that learns from partial results to several competing objectives: dense and crack-free deposits, strength, ductility, and resistance to corrosion in seawater.
2. Extend it to many experiments in progress at once, with results that return at different times.
3. Give it physics computed for the process, thermodynamic calculations evaluated along the thermal history of a DED deposit, and measure what that physics is worth.
4. Demonstrate the method on a physical workflow that designs copper alloy powders for blown-powder DED, from powder to deposit, repair coupon, tensile test, and seawater corrosion test. Test whether its stated confidence in a candidate matches what the deposits and tests later show.
5. Measure how well a one-hour electrochemical screen predicts immersion tests that take weeks, so that the long tests run only where they can change a decision.

### 1.2 Copper alloys in naval seawater service

Navy specifications cover copper-nickel tube for the piping systems of surface ships and submarines [2] and NAB castings for seawater service [3]. NAB is widely used for valve bodies, pump casings, and impellers, with a large market in submarine fleets [4]. Copper-nickel owes its resistance to a thin, adherent film of corrosion products, mainly cuprous oxide with nickel and iron. The film builds up over weeks to months of exposure [13]. In one study the films on 90-10 tubes in natural seawater were still changing after six months [14]. The iron content of the alloy changes the makeup of the film [20]. NAB is a cast alloy of copper with about 9% aluminum and 4 to 5% each of iron and nickel (by weight). Its microstructure of $\alpha$ grains and several intermetallic $\kappa$ phases resists seawater well, but the $\alpha$ phase in the lamellar $\alpha + \kappa$ eutectoid, and any retained $\beta$ phase, corrode selectively, above all in crevices [21, 22]. Navy NAB castings therefore get a temper anneal at $1250 \pm 50\,^{\circ}\mathrm{F}$ (about $677\,^{\circ}\mathrm{C}$) for at least 6 hours, propellers excepted [3], which improves their corrosion resistance [4].

Printing changes these microstructures, and printing changes how alloys corrode [23]. How NAB is processed sets its microstructure [24], and NAB has now been printed by laser powder bed fusion [25], wire-arc deposition [26, 27], and laser DED [7, 28]. Laser-deposited NAB contains martensitic $\beta'$ along with $\alpha$ and $\kappa$ phases [7, 28] and is much stronger than cast NAB [28]. Its corrosion depends on the process. NAB from laser powder bed fusion formed stronger films than wrought NAB but resisted localized corrosion worse, and annealing reversed both [29]. Wire-arc

NAB corroded in its residual $\beta$ phase [30]. Whether annealing helps depends on the starting microstructure. After 480 hours in salt solution, the same 700 °C anneal had lowered the corrosion current of cast NAB and raised that of sintered and hot isostatically pressed NAB, each about threefold [31]. The authors call the differences small. As-built wire-arc NAB resisted corrosion better than annealed coupons, in a test with unequal immersion times [32]. Laser-deposited NAB on stainless steel was attacked to a depth of 3.4 $\mu$m in 21 days in salt solution where iron from the steel had diluted it, and 1.8 $\mu$m over a nickel-chromium interlayer [33]. In welded NAB, retained $\beta'$ in the heat-affected zone corroded selectively in seawater [8], a warning for repairs. Laser cladding of cast NAB has used an NAB consumable [34], and DED repair of cast NAB a low-nickel copper alloy powder [35, 36]. That powder was later blended in the melt pool with an aluminum alloy powder to vary the aluminum content [37].

In wire-arc 70-30 copper-nickel, the dendrite cores hold more nickel than the regions between them, and those regions corrode first [38]. The corrosion of powder-bed 70-30 has also been studied [39], and its resistance depends on the laser settings [40]. At their best settings, both wire-arc and powder-bed 70-30 have matched or beaten wrought plate in electrochemical tests in salt solution [38, 40]. Laser cladding of Cu-30Ni dates to 2005 [41], and later blown-powder DED of copper-nickel has mixed copper and nickel-alloy powders in the melt pool to make graded layers and thermoelectric alloys [42, 43, 44]. The one corrosion study we found of copper-nickel deposited by blown-powder laser DED is from 2011: Cu-30Ni made by $CO_2$-laser direct metal deposition on wrought 70-30 plate [45]. A later review reports that the deposit corroded faster than the plate and attributes this to its porosity [46] [TODO Sterling: check the 2011 article before citing a direction. Read literally, its abstract ranks the deposit as the most corrosion resistant and the plate as the least, the reverse of the review. A secondary source found by web search quotes ASTM G31 rates of 0.36 mpy for the deposit and 0.08 mpy for the plate. Neither Edison nor this runner could open the article]. NAVSEA has written specifications for 70-30 copper-nickel made by wire-arc DED and by laser powder bed fusion, and for NAB made by wire-arc DED. The wire-arc ones are alternatives to castings, and all three keep the existing compositions [9, 10, 11]. We found none for either alloy made by blown-powder laser DED. In 2023 a Navy official said, as Defense News reported, that for naval metals "including copper-nickel," "more research is needed" on how printing "affects the metals' fatigue and corrosion properties" [47]. A composition chosen for casting is not necessarily the best one for DED, and each composition and set of deposition settings needs its own tests.

Corrosion tests of copper alloys follow the same cost pattern as the rest of the workflow. Open-circuit potential, linear polarization, and impedance measurements in substitute ocean water [48] take about an hour and give a corrosion rate and a measure of the surface film [49, 50]. Immersion tests with mass loss [51, 52] and cross-sections that measure the depth of selective attack take weeks. Tests in flowing seawater, which set the velocity limits used in piping design, take months. The search can therefore treat corrosion the way it treats cracking and strength: a cheap screen for many candidates and long tests for a few. The cheap screen can mislead, though. An hour-long measurement on a fresh surface sees neither the film that forms over weeks on copper-nickel nor slow selective attack in NAB. On wrought 70-30 copper-nickel in synthetic seawater, the corrosion current fell more than eightfold over the first ten days [53], and welded NAB corroded faster in natural seawater than in simulated seawater [54]. In a study with NSWC Carderock, Volta potentials measured in air on NAB plate ranked its phases backwards: the phase with the highest potential was the one attacked in solution [55]. Knowing when the screen can be trusted is part of the research.

Supply and melting add one more objective. Nickel has been on the U.S. list of critical minerals since 2022, and copper joined it in 2025 [56, 57]. Nickel also raises the melting range: 70-30 copper-nickel melts between about 1,171 and 1,238 °C, 90-10 between 1,099 and 1,149 °C, and NAB between 1,043 and 1,060 °C [58]. A replacement for 70-30 copper-nickel that needs less nickel would use less of a critical mineral and would be easier to atomize and deposit. The search will carry nickel content as an objective to minimize where the corrosion trade-offs allow.

### 1.3 Scientific concepts

The search method treats the manufacturing workflow as a network of linked models [59]. Each step gets its own statistical model (a Gaussian process), which predicts the outputs of that step and how uncertain they are. The models are linked in the same order as the steps (Fig. 1), so the outputs of one step are inputs to the next and uncertainty carries through the whole network. Because each step has its own model, the network learns from partial results, and a candidate can stop at any step. Buathong et al. added a cost-aware rule to this network [60]. For each possible experiment, the rule estimates how much the result is expected to improve the search, divides that by the cost of the experiment, and runs the best one. In this project the deposition step is itself split in two by cost. A small DED system at the University of Utah, 45 miles from BYU, deposits tracks, walls, and small blocks from grams to half a kilogram of powder on a day trip. A larger one at the Colorado School of Mines, 500 miles away, builds tensile, corrosion, and repair specimens from 2 kg lots, with results in weeks. Our rule treats the two as linked steps with their own costs and pays for the larger one only where the smaller one shows that a composition could change the outcome (Section 1.8.2).

The physics comes first, in the manner of integrated computational materials engineering (ICME) [61, 62]: calculations decide where experiments go, and experiments correct the calculations. Calculation of phase diagrams (CALPHAD) in Thermo-Calc and pycalphad [63, 64] gives the liquidus and freezing range of each composition and an index of how prone it is to cracking as it solidifies [65]. Such indices have been checked against printed aluminum alloys, steels, and nickel superalloys, but not copper alloys [66]. It also gives how nickel, iron, and aluminum divide between the dendrite cores and the last liquid to freeze, and the phases stable at each temperature, including the $\beta$ and $\kappa$ phases of the bronzes. DED adds a thermal history. For each set of deposition settings, a moving point-source model of heat flow [67, 68] gives the cooling rate through solidification and the peak temperatures that later passes impose on each layer. We calibrate it to measured melt-pool sizes and give it the temperature-dependent properties that NSWC Carderock has measured for 70-30 and 90-10 copper-nickel and for NAB plate and wire-arc NAB [69, 70, 71, 72]. Room-temperature constants in their place can reverse the predicted distortion of a NAB deposit [73]. For new compositions, which these databases do not cover, we estimate the properties from the nearest measured alloy and CALPHAD and carry the added uncertainty into the model. The CALPHAD calculations are then evaluated at those temperatures, to estimate which phases can form as the deposit solidifies and which can form or dissolve as it is reheated. These outputs enter the statistical models as inputs with learned weights, so the search can discount any of them that proves uninformative. The calculations run continuously on BYU high-performance computing, and each new result is available to the next decision.

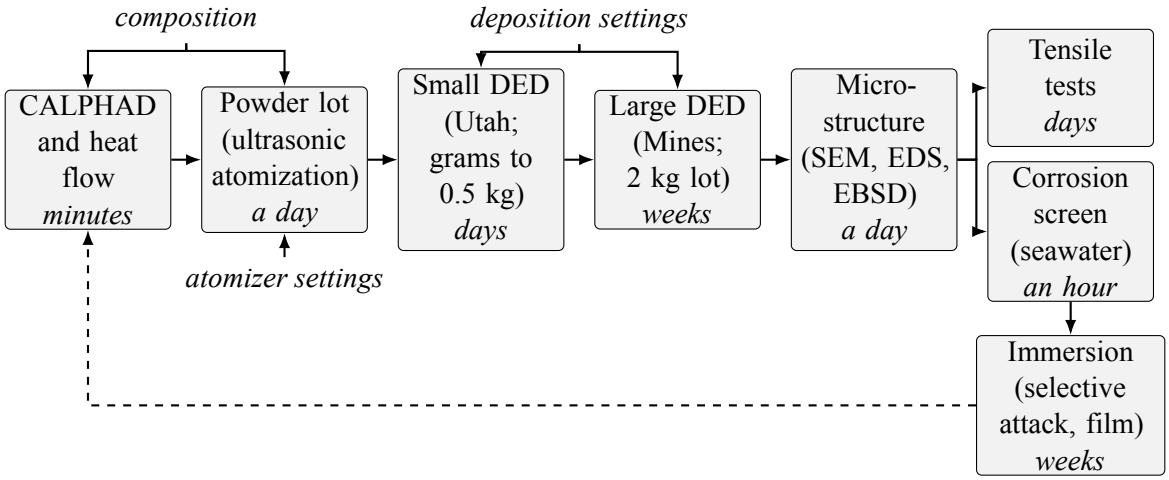



**Figure 1.** The workflow as linked models, with typical times per result in italics. DED is bought at two facilities that the search treats as two steps of different cost. The small system at Utah melts pre-placed tracks from grams of a first powder lot, then blown-powder tracks, walls, and small blocks from about 0.5 kg. A composition that earns it is scaled up to the 2 kg lot that the hoppers of the large system at Mines need, for tensile, corrosion, and repair specimens. That takes weeks, including the extra atomizer runs and turnaround at the facility. Composition enters at the first two steps, atomizer settings at the second, and deposition settings (laser power, travel speed, and for blown powder the feed rate) at the third and fourth. The heat flow model also takes the deposition settings. After microscopy the network branches to tensile tests and to corrosion tests. Dashed arrow: each result updates the models, which choose the next experiments and steps by expected gain per cost.

### 1.4 Comparison with other approaches

To our knowledge, no cost-aware search over linked models that can stop a candidate at any step has been extended to several objectives or to results that arrive out of order. None has been demonstrated on a physical alloy workflow. Published tests of the linked-model methods use test functions and simulators [59, 60]. Multi-stage Bayesian optimization for self-driving laboratories was tested on synthetic and past data with one objective [74]. Cost-aware batch optimization of refractory alloys treats measurements of unequal cost as separate sources rather than linked steps, in a computational campaign [75]. Closed-loop campaigns on printed alloys use one model and a fixed set of measurements [76]. A search guided by model uncertainty has chosen compositions of multi-principal element alloys deposited by DED, scored on hardness [77]. One model has combined a cheap analytical heat-flow model with costlier finite-element simulations to map DED settings [78]. Neither linked the steps or their costs. Alloys designed for printing by computation, such as an oxide-dispersion-strengthened NiCoCr alloy [79] and high-strength aluminum alloys [80, 81], were designed and then printed as single compositions. An ONR-funded CALPHAD-based ICME framework screened 450,000 compositions of a naval steel for the chance of a successful print under the scatter of powder composition [82]. It worked from composition and heat treatment, without the thermal history of the build, corrosion, or a choice of which experiment to run. A nickel superalloy has been designed with a neural network for laser repair [83]. Machine learning has refined powder-bed settings for 70-30 copper-nickel at the Naval Postgraduate School [84], and Bayesian optimization has tuned a gas atomizer [85]. Both tune the process for one fixed

alloy. Standard Bayesian optimization with several objectives handles competing properties and several experiments at once [86] but treats the whole chain as one costly experiment. Printed copper alloys for seawater have used the cast or wrought compositions [9, 10, 11], and their corrosion studies characterize one alloy and one process at a time [29, 30, 38, 40]. A 2026 study adjusted the composition of NAB for laser additive manufacturing to improve its as-deposited strength and corrosion resistance [12], and adding 0.1% titanium to wire-arc NAB lowered its corrosion current by 16% [87]. Both show the payoff of designing for the process. Autonomous and high-throughput corrosion work has used a scanning droplet cell to screen coatings and has run fast screens before slower, more thorough tests [88, 89], but without deciding, candidate by candidate, when a long test is worth its cost. A recent perspective sketches, as a hypothetical case, an autonomous laboratory that would design a naval brass by DED, with a hardness screen ahead of erosion-corrosion tests [90]. This project combines the linked network, the cost per step, physics computed for the process, several objectives including corrosion, and staggered results. We test the combination on real deposits, with cheap and costly measurements paired on the same candidates and an audit of what the screens discard.

### 1.5 Preliminary results

**Powder to part.** Since the lab started at BYU in December 2025, we have set up a CALPHAD workflow on BYU high-performance computing, installed the ultrasonic atomizer, and completed a powder-to-part demonstration on a baseline aluminum alloy. Atomized powder, laser powder bed fusion builds, scanning electron microscopy (SEM) and energy-dispersive X-ray spectroscopy (EDS) results, and tensile data are in hand. The induction module of the atomizer also melts copper alloys, up to its working limit of 1,300 °C. NAB and 90-10 copper-nickel melt about 250 and 150 °C below that limit, and 70-30 copper-nickel within about 60 °C of it. [TODO Sterling: add the first copper powder (SEM image and size distribution) from the runs planned for the week of 19 October; attachments can be replaced until SPO submits on 28 October.]

**Powder doser.** We have built an open-source auger powder doser that weighs each dose on an analytical balance (0.1 mg). A Bayesian optimization campaign tuned its eight settings unattended in under 2.5 hours (Fig. 2). With the recommended settings it dispenses 0.5 g of salt in under two minutes, within 3 mg, where hand-tuned settings took three to six minutes. In October 2026 it began dosing atomizer charges, Al 4047 powder into aluminum cups for a composition series. The 15 wt% cup ended at 6.5676 g against a 6.5674 g target. Across twelve powders, the feed per auger turn fell as the reported angle of repose rose (Spearman $\rho = -0.84$), so its dosing record may also serve as a cheap reading of how a lot flows. Task 2 uses it the same way, for the powdered additions to each copper charge.

**CALPHAD for copper alloys.** We already run the CALPHAD step on copper, with pycalphad [64] and an open database that covers copper, nickel, aluminum, iron, and manganese [91]. The equilibrium solidus and liquidus of 70-30 copper-nickel come out at 1,171 and 1,229 °C, against handbook values of 1,171 and 1,238 °C [58], and those of 90-10 within 18 °C (Fig. 3). Each composition takes about a second on one processor, so a 42-point map of the nickel and aluminum ranges took under a minute. It shows the liquidus passing the 1,250 °C cap of Table 1 only where nickel and aluminum are both high. For NAB the database places the solidus within 7 °C of the handbook value and predicts that $\beta$ is gone below about 730 °C at equilibrium. Any $\beta$ left in a deposit is therefore set by the cooling rate, the effect that the process-aware inputs of Advance 3 compute. Its NAB liquidus,

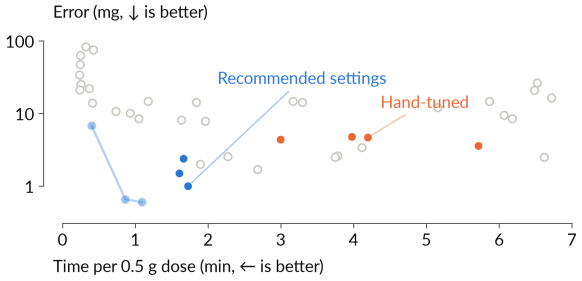



**Figure 2.** Bayesian optimization of the powder doser, on salt. Each gray circle is one 0.5 g dose of the campaign, plotted by time and error. Light blue joins the best single doses, blue marks doses at the recommended settings, and orange marks hand-tuned settings.

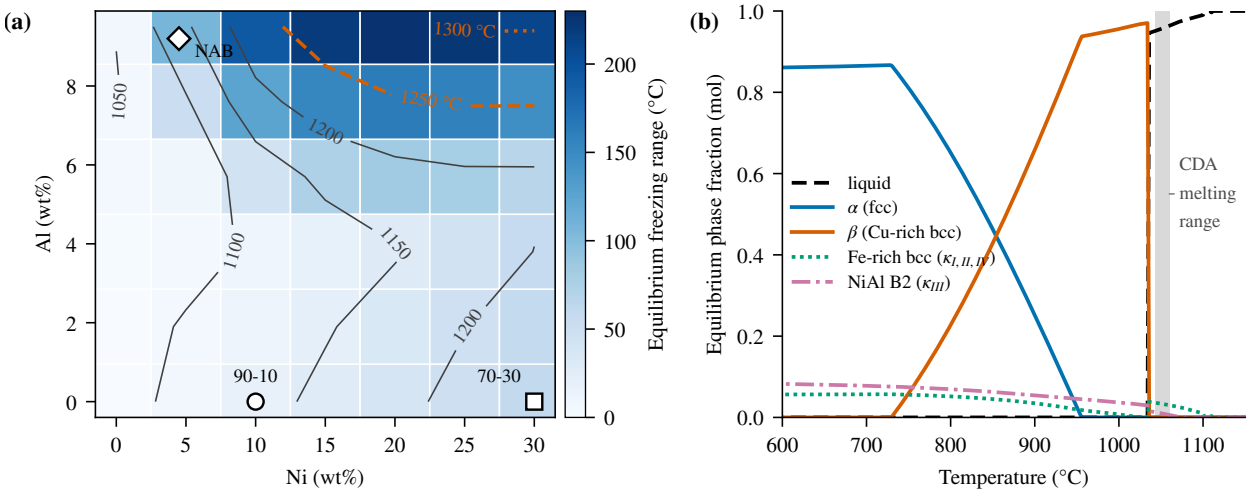



**Figure 3.** pycalphad calculations with an open database, mc_ni from MatCalc [91], which covers Cu-Ni-Al-Fe-Mn but was built for nickel alloys. (a) Equilibrium freezing range (color) and liquidus (contours, °C) over the nickel and aluminum ranges of Table 1, with 1 wt% iron. Orange lines mark the 1,250 °C liquidus cap and the 1,300 °C limit of the atomizer, and symbols mark the reference alloys. Values in the high-nickel, high-aluminum corner are upper bounds. (b) Equilibrium phase fractions of NAB (C95800) against temperature. The gray band is the handbook melting range [58].

though, is 51 °C too high, so open calculations will serve as a check on the Thermo-Calc copper database [92].

**Deposition.** Mines has already melted copper-nickel powder spread in grooves, the pre-placed tracks that start Task 2 [TODO Sterling: ask Sam whether these tracks, and a photo, may be shown]. At Utah, the Fang group installed its MX-Lab to develop new alloys, feeding up to six powders at once [93].

We also simulated the cost-aware method on a synthetic version of the workflow (Figs. 4 and 5), made for the aluminum work. With the crack screen ignoring zirconium, screening and then tuning the process stalled at 63% of the optimum after $100,000. The cost-aware search, and a simpler one that reused powders, reached that level after about $26,500 (3.8 times less spend; 90% confidence interval 2.0 to 5.7). Searches that ran every candidate through the full chain caught up later, and screening kept pace when its physics was right. The simulation is hand-built and has no measurement noise, with one objective, no corrosion branch, no process physics, and one experiment at a time, so it does not test the advances proposed here. It does show that letting the search choose the step can pay off when the physics model is incomplete. The DED workflow has the same structure: a physics screen that can miss effects, a powder step, a deposition step, and tests.

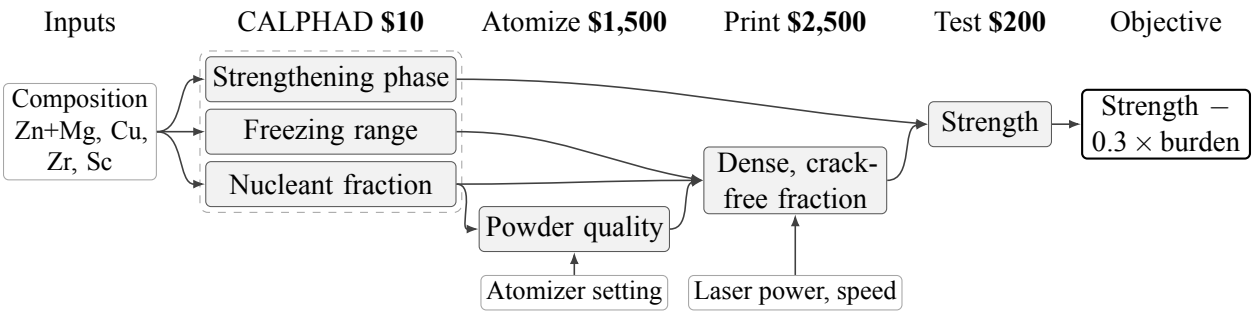



**Figure 4.** Synthetic workflow used for Fig. 5, built for the aluminum version of this workflow: a hand-written function for each box and a rough total cost (time, materials, supplies) for each step. Burden is a penalty on scarce elements such as scandium. The nucleant (Zr, Sc) cuts cracking, an effect the CALPHAD crack screen does not see. Task 1 replaces these functions with copper and DED ones.

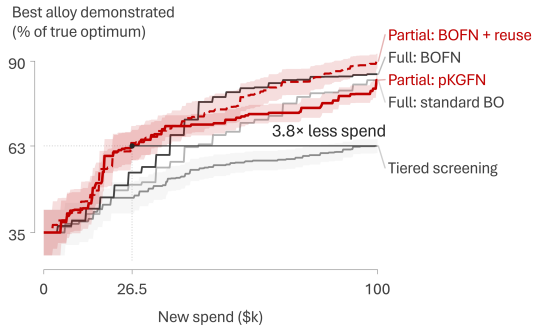



**Figure 5.** Best result found versus spend in simulations where the CALPHAD crack screen ignores zirconium (means of 20 campaigns, ±1 standard error). Red searches choose the next step. Gray ones follow a fixed order. pKGFN is the cost-aware method [60] and BOFN the linked-model method [59]. Reuse reprints earlier powders.

### 1.6 Method advances

**Advance 1: several competing objectives.** The published cost-aware method optimizes a single objective. A copper alloy for DED repair must balance deposit quality (dense, crack-free, and sound at the fusion line with the part it repairs), strength, ductility, and corrosion resistance in seawater, with no weighting known in advance. We will extend the cost-aware rule to several objectives. We will replace the expected improvement in one objective with the expected growth of the set of best trade-offs (the Pareto front), divided by cost as before. That growth is measured by the volume of property space that the trade-offs outperform [86]. The difficulty is that each cheap measurement informs some objectives strongly and others weakly. Single tracks show cracking and dilution well but say little about strength or corrosion, and a polarization measurement says much about corrosion rate and nothing about strength. The rule has to account for that through the network. We will derive the rule, check it against exhaustive calculations on small problems, and measure its computing time on BYU high-performance computing.

**Advance 2: results that arrive out of order.** The published method chooses one experiment at a time and waits for its result. In this workflow a DED session deposits many specimens, microscopy results arrive days later, immersion tests run for weeks, and thermodynamic calculations finish in minutes. We will extend the rule to groups of experiments whose results arrive at different times, so that new experiments are chosen while others are still running. Experiments in progress will enter the models as stand-in results drawn from the current predictions, a standard way to avoid choosing the same experiment twice [94]. We will also test how stable the choices are when a

late result contradicts early ones. For example, a block that looked dense on its surface may show cracks under the microscope, or an alloy that passed the polarization screen may show selective attack after a month of immersion.

**Advance 3: physics computed for the process.** Standard CALPHAD inputs describe equilibrium, or solidification without diffusion in the solid, and ignore the thermal history of the process [95]. We will add inputs computed along the DED thermal history (Section 1.3): cooling rate through solidification, the peak temperatures of reheating, and the phases that CALPHAD allows at those temperatures. Hypothesis H3 tests whether these inputs make the models' predictions more reliable than equilibrium inputs, and whether equilibrium inputs beat data alone. The answer measures what incomplete, process-aware physics is worth inside a search, which sets how much simulation a campaign should buy before it buys experiments.

**Advance 4: a physical workflow from powder to repair.** We will demonstrate the advances on a physical workflow (Fig. 1) in which the search chooses the composition, the atomizer settings, and the deposition settings, and decides at each step whether a candidate goes on. After microscopy, specimens go to tensile tests, or to the corrosion screen and, for selected candidates, to immersion tests. Candidates that deposit well get a repair coupon: a deposit on cast NAB or wrought copper-nickel plate, checked for cracks, dilution, and hardness across the fusion line. These checks do not show whether a repair will last. Crack growth in printed steels varies with the build process [96], so fatigue and residual stress of repairs are left to Navy laboratories. Pre-placed tracks on the same plates, at Utah, come first. The demonstration includes real deposit-to-deposit scatter, outputs computed from microscopy images, and three inputs that enter at different steps.

**Advance 5: checking what the screens discard.** A cheap screen saves money only if it does not discard good candidates, and Section 1.2 shows that a short corrosion screen can also pass bad ones. Each round, the search will send a small random share of the candidates it would stop, about one in ten, on to the next step anyway. These audited candidates measure how often each screen is wrong, and they keep the models from learning only from candidates that the screens favored.

### 1.7 Hypotheses and rejection criteria

**H1 (cost).** The cost-aware search, with several objectives and experiments in progress, will reach a target set of trade-offs at lower total cost than standard Bayesian optimization with several objectives, which carries every candidate through the full chain. H1 will be rejected if, across matched campaigns with the same budget, the median cost to reach the target is not lower than that of the full-chain baseline.

**H2 (calibration).** The predicted probabilities that a candidate will deposit dense and crack-free, fuse to the repair substrate without cracks, and stay close to its intended composition will match the observed rates on deposits the models have not seen. H2 will be rejected if observed success rates differ from predicted probabilities by more than 10 percentage points.

**H3 (process-aware physics).** CALPHAD outputs evaluated along the DED thermal history will improve that calibration over CALPHAD outputs at equilibrium, which will in turn improve it over models trained on data alone. H3 will be rejected if the process-aware inputs do not reduce calibration error on unseen deposits compared with the equilibrium inputs.

**H4 (pending results).** Choosing new experiments while others are pending will reach the target in less calendar time than waiting for each batch to finish, at equal spend. H4 will be rejected if, in median calendar time, the version that works around pending results is no faster than the version that waits.

**H5 (corrosion screen).** Corrosion rates, breakdown potentials, and impedance from the one-hour screen, together with composition and microstructure, will predict which candidates pass a 30-day immersion test. To pass, a candidate needs a mass loss no greater than that of the 70-30 copper-nickel reference and selective attack no deeper than [TODO Sterling: depth. Immersions of 21 and 30 days in salt solution found at most 1.8 to 3.4 $\mu$m in laser-deposited NAB [33] and more than 3 $\mu$m in cast NAB [97]. Options: a few micrometers, or no deeper than in the cast NAB reference]. H5 will be rejected if, among tested candidates (audited ones included), those that the model ranks in the top half pass no more often than those in the bottom half. We will also report the rank agreement of screen and immersion and the share of passing candidates that the screen rejects, with confidence intervals, and will test a few candidates just either side of the cutoff.

We will fix and post the targets, baselines, budgets, and number of matched campaigns before each test starts [TODO Sterling: confirm a public pre-registration site, e.g., OSF]. We test H1 and H4 first on simulated benchmarks and on benchmarks built from past data, where many matched campaigns are affordable, and then in live campaigns, where only a few are.

### 1.8 Research plan

**Task 1: methods (Ph.D. student 1).** Derive and implement the rule for several objectives (Advance 1), the rule for pending results (Advance 2), the process-aware physics inputs (Advance 3), and the audit of screened-out candidates (Advance 5). Build simulated benchmarks that replace the functions of Fig. 4 with copper and DED ones, with noise, several objectives, a corrosion branch, and delays drawn from measured turnaround times. Build benchmarks from past data: our aluminum data and the copper data that Task 2 adds each year.

**Task 2: experimental platform (Ph.D. student 2).** We will make powder for each candidate in our laboratory. Oxygen-free copper bar is machined into cups that hold the charge. We weigh in nickel, iron, manganese, aluminum, and other additions to 0.1 mg as pure metals or copper master alloys. A charge is about 0.5 kg, as in the published CuSn8 runs [98] [TODO Sterling: confirm the charge size after the first copper runs]. Minor additions that come as powder are dispensed into the cup by our powder doser (Section 1.5), which weighs and logs each dose, so every lot carries a record of what went into it. The charge is melted in a graphite crucible in the induction module of the ultrasonic atomizer (AMAZEMET rePOWDER) and poured through a 0.5 mm nozzle onto a vibrating plate, where it breaks into droplets. Nickel and iron react with graphite. Before melts that contain them, we spray the crucible with a thin boron nitride coating, which lasts a few runs, and pour through a boron nitride nozzle, as the manufacturer advises. NAB has already been atomized from a graphite crucible on a rePOWDER [17]. Many alloys wet bare carbon-carbon composite poorly [98], so the plate is carbon-carbon with a molybdenum- or tungsten-based coating where the melt lands. The first part of each pour heats the plate until the melt wets it, and atomization is then continuous. With such plates on our model of atomizer, the manufacturer turned 80 to 90% of each charge of a tin bronze (CuSn8) into powder [98]. [TODO Sterling: confirm with AMAZEMET the plate and nozzle for NAB and copper-nickel; the published NAB runs used a bare carbon-fiber plate

that left fibers in the powder [18]]. The median droplet size follows the wavelength of capillary waves on the plate, so it falls as the frequency to the power $-2/3$ [99]. The plates on our unit vibrate at 40 kHz, where the Lang relation gives a median near 47 $\mu$m for a copper melt. Measured powders are coarser. NAB atomized at 40 kHz had a median of 98 $\mu$m [18], about twice that estimate, and much of the CuSn8 powder made at 40 kHz on our model fell between 63 and 150 $\mu$m [98]. Both suit the DED system described below, which takes powder of about 50 to 150 $\mu$m, sometimes up to 200 $\mu$m, and is usually fed cuts of 45 to 105 or 100 to 200 $\mu$m. Published blown-powder work on bronzes used 45 to 106 and 45 to 150 $\mu$m cuts [28, 100]. We sieve each lot into those two cuts. We check its size distribution (sieve analysis and SEM images), flow and apparent density with a Hall funnel [101, 102], shape, satellites, and stray plate fibers (SEM), and oxygen and carbon content. [TODO Sterling: name the service laboratory: inert-gas fusion for oxygen, combustion for carbon]. Powder oxygen also matters at a repair. In a model system, additions whose oxides are more stable than that of the parent metal improved wetting and bonded area [103], so the models relate the oxygen of each lot to fusion on cast NAB and wrought copper-nickel. The atomizer settings the search chooses are the melt temperature, the pour pressures, and the plate amplitude, and the share of each lot in each cut is an output of the powder step. With the gas flows and the laser spot, the size distribution sets how much of the powder the melt pool catches [15]. The search chooses the atomizer settings along with the composition, so the powder, not only the alloy, is designed for blown powder.

We buy DED as machine time at two facilities, and the search treats them as two steps of different cost (Fig. 1). Neither facility is a subawardee. For the small step we use an InssTek MX-Lab at the University of Utah, in the laboratory of Professor Z. Zak Fang, 45 miles from BYU, which his group has offered to the project. It has a 500 W fiber laser, 150 mm of travel on each axis, control of melt-pool height, and six powder feeders that can each be run separately [104, 93]. [TODO Sterling: confirm with Fang: the rate, the smallest powder charge per feeder, the argon chamber and any preheat, and that a BYU student runs it after training]. We start every composition here. We spread powder from a first lot in grooves machined in plates, including cast NAB and wrought copper-nickel plate, and melt it at several laser powers and speeds. These pre-placed tracks use a few grams of powder each and show cracking, porosity, dilution, and fusion to the repair substrates. They differ from blown-powder tracks in heat input and in how the powder reaches the melt pool, so the models treat them as a cheaper, less faithful measurement. Once its tracks are sound, a composition gets blown-powder deposits from one 0.5 kg lot. Single tracks at several laser powers, travel speeds, and powder feed rates show melt-pool size, dilution, cracking, and the share of the powder caught in the melt pool. Thin walls show cracking and heat buildup, and small blocks show density and porosity. The heat flow model is calibrated separately for each kind of track, and the audit of Advance 5 measures how often the tracks mislead.

For the large step we buy sessions on a FormAlloy X2 blown-powder laser DED system at the Colorado School of Mines, run by facility staff at its external rates for machine time and labor. Facility staff plan 3 hours of machine time and 2 hours of labor for a typical session of four 25 mm cubes. Mines plans on the melt pool catching about half the powder fed, and needs about 2 kg of each powder to fill the hoppers of the X2. Since 2 kg takes about six atomizer runs, only compositions whose small blocks at Utah reach 99% relative density are scaled up. Their sessions at Mines build blocks for sub-size tensile bars (ASTM E8/E8M) and corrosion coupons at the chosen settings. They also build repair coupons on thick cast NAB and wrought copper-nickel plate (Advance 4), where fusing

the first layer to a large heat sink needs more laser power than the MX-Lab has. [TODO Sterling: confirm the X2's laser (wavelength, power, spot size), its shielding, substrate preheat, turnaround, and that powder left in the hopper comes back to us]. Commercial 70-30 copper-nickel and NAB powders, deposited at both facilities, are the baselines [TODO Sterling: confirm commercial powder sources, at least 2 kg of each; Mines can also supply powder at cost]. The search decides which compositions go to Mines at all.

At BYU we will examine selected deposits by SEM, and by EDS and inductively coupled plasma mass spectrometry for composition, since manganese and aluminum can be lost to evaporation and oxidation [95]. Electron backscatter diffraction (EBSD) gives grain size, texture, and phase fractions, including martensitic $\beta'$ in the bronzes. Finalists also get transmission electron microscopy, so that fine $\kappa$ precipitates, which SEM can miss, are measured rather than assumed. Replicate tracks and blocks at fixed settings will measure deposit-to-deposit scatter. Image analysis scripts will turn micrographs into porosity, crack length, dilution, grain size, and phase fractions with stated uncertainty.

Corrosion tests follow the same staging. The screen runs on the liquid-handling gantry in our laboratory (CubXL, Ursa Laboratories), which is already driven by scripts, so the search can schedule it like any other step. The gantry fills small O-ring wells, 3 to 4 mm across, clamped on polished faces, with substitute ocean water [48], and a potentiostat dedicated to the loop measures open-circuit potential, linear polarization, impedance, and cyclic polarization [49, 105]. Small cells can add crevice attack at the seal and shift breakdown potentials [106]. We therefore photograph each spot before and after and check the wells against a conventional 1 $cm^2$ flat cell on the references. [TODO Sterling: choose the potentiostat: a module unit with impedance on the gantry, cross-checked on a borrowed laboratory instrument, or the benchtop unit and flat cell in Field 11; see cubxl-electrochemistry.md]. Several wells fit on one specimen, so a repair coupon is measured in the deposit, at the fusion line, and in the heat-affected zone of the substrate, where welded NAB was attacked selectively [8]. The gantry then pipettes the used electrolyte of each well into a tube for inductively coupled plasma mass spectrometry, which shows which elements left the surface, for example aluminum from NAB or nickel from copper-nickel. ASTM G61 is written for iron-, nickel-, and cobalt-based alloys, so we will report how its procedure behaves on deposited copper alloys. We measure wrought 70-30 copper-nickel and heat-treated cast NAB as references in every round. The screen is a ranking that has to be calibrated, so deposits and references get the same preparation and exposure, in replicate, and the conditions that control seawater tests are recorded [107, 108]. Candidates that the search selects get 30-day immersion tests in capped vials on the deck of the gantry. The gantry draws a few microliters of each solution at about 1, 3, 10, and 30 days for the same analysis. The metal released over time then enters the models as an intermediate result between the one-hour screen and the final mass loss. We weigh the coupons before and after [51, 52] and section them to find the deepest selective attack. Repair coupons are also immersed whole, with a stated ratio of deposit to substrate area, because a repair is a galvanic couple [109]. Simulated pits in printed microstructures grow deeper than in a uniform model [110], and crevice damage in printed stainless steel, measured in three dimensions, has tracked the electrochemistry [111]. Microstructure therefore stays in the corrosion model. The leading candidates get crevice and longer immersion tests [TODO Sterling: confirm the durations, and natural or substitute seawater]. Tests in flowing seawater, which take months, are left to Navy laboratories, such as the marine corrosion laboratories at NSWC Carderock [112], for the three leading candidates

and the two references [TODO Sterling: confirm the number with Carderock] (Section 1.10).

We chose the search space the way we chose our aluminum palette: a literature review of each candidate element, then a check of its coverage in the CALPHAD database. We reviewed fifty elements from the periodic table one at a time. They include every element that appears in a commercial copper alloy, a copper alloy or welding filler specification, a deoxidizing practice, or the crucible and plate materials of the atomizer. Each review asked four questions. Does the element help seawater resistance, deposit soundness, or repair? What sets its upper limit? Can it be melted and atomized below 1,300 °C without attacking the crucible or plate? Does the Thermo-Calc copper database (TCCU7 [92]) assess its systems? We search eight elements from the start (Table 1), with copper as the balance at no less than 65% and a calculated liquidus no higher than 1,250 °C. In Year 2 we will test zirconium and boron, tin and germanium in NAB, and one rare earth, lanthanum or cerium, which improved a 90-10 alloy in salt solution [113]. Rules tied to the aluminum content keep each candidate near known metallurgy. Above 7% aluminum, nickel must at least equal iron, as it does in the nominal composition of Navy NAB castings [3]. Below 1.5%, iron stays at or below 2.3%, the upper limit of the iron-rich 70-30 grade C71640, so that it remains in solid solution [114]. Chromium and niobium are used only where aluminum is at most 3%, and silicon above 0.1% only with niobium, as in cast 70-30 copper-nickel [115]. Retained $\beta$ is not limited by a fixed formula. The process-aware CALPHAD inputs predict it (Advance 3). The space includes the region between the two families, copper-nickel-aluminum alloys such as the high-strength MARINEL [116], where low-nickel candidates may lie. Beryllium and cadmium (toxic powder and fume), zinc (it evaporates in the melt pool and dezincifies in seawater), and the refractory metals (immiscible, and far above the melting limit) are excluded. Lead, sulfur, phosphorus, carbon, bismuth, and the other impurities are held at or below the welding-grade limits for copper-nickel [117]. All the searched elements but iron are on the 2025 U.S. list of critical minerals [57]. Nickel dominates by mass and is carried as an objective.

**Task 3: live campaigns and hypothesis tests (both students).** Run live campaigns in which the search chooses each experiment and step. The campaigns are supervised by people, who book the DED sessions and handle the coupons. Compare the search on equal budgets with random selection, with a fixed funnel that passes the top candidates at each step, and with standard Bayesian optimization with several objectives over the full chain. Versions that drop the costs or the physics inputs show what each adds. A funnel that looks sensible can do worse than random selection [119]. Score each on total dollars and DED hours to a confirmed candidate. Report the false rejections at each step, the calibration at each link, and the error on compositions outside the training family, such as NAB predicted from copper-nickel data. We test the transfer of results from the Utah system to Mines on paired compositions, as has been done between wire and powder DED [120]. Test H2 and H3 on unseen deposits, H5 on the audited and selected candidates, and H1 and H4 in matched campaigns.

**Task 4: release and transition.** Release the software, data, and models each year (Section 5). Write a short guide for applying the method to other alloys, deposition processes, and test sequences, aimed at Navy laboratory and depot engineers. The search narrows the field. Each finalist then gets replicated builds and a dataset in the form of the Carderock property databases [69, 71]: powder specification, deposition settings, and tensile, corrosion, and repair-coupon data, with the raw files public. We share them with materials engineers at NSWC Carderock, who can decide which can-

**Table 1.** Elements searched from the start (wt%), with copper as the balance (at least 65%) and a calculated liquidus of at most 1,250 °C. "Cu-Ni only": used where aluminum is at most 3% (silicon: below 7%).

| Element | Range | Role | Upper limit set by |
|---|---|---|---|
| Ni | 0–30 | Protective film in Cu-Ni; $\kappa$ phases in NAB | Liquidus near the atomizer's limit; supply (an objective) |
| Al | 0–9.5 | Aluminum bronze; Cu-Ni-Al alloys | Retained $\beta'$ martensite as deposited |
| Fe | 0–4.5 | Film in Cu-Ni; $\kappa$ phases in NAB | Iron out of solution in Cu-Ni; coarse $\kappa$ in NAB |
| Mn | 0–4 | Deoxidizer; partial nickel substitute | Evaporation and oxidation in melting and DED |
| Cr | 0–0.7, Cu-Ni only | Resistance to fast flow [118] | Cr-rich particles; crevice attack in still water |
| Si | 0–0.5, Cu-Ni only | Deoxidizer; strength with Nb | Silicides; weld cracking |
| Nb | 0–0.8, Cu-Ni only | Weldability of cast Cu-Ni | Undissolved Nb-rich particles |
| Ti | 0–0.5 | Deoxidizer in Cu-Ni welding filler [117] | Oxide and nitride inclusions; crucible attack |

didates merit flowing-seawater and service tests. The finalists will be candidates for Navy qualification testing. Component tests, fatigue, inspection methods, and design allowables would remain, as would the decision whether a controlled repair consumable or a revision of a specification such as MIL-PRF-32816 [9] is warranted.

### 1.9 Expected results and new knowledge

If successful, the project will produce:

1. The first cost-aware method of this kind for competing properties and staggered results.
2. A measured answer to how much a staged, cost-aware search saves over full-chain testing on a real alloy workflow.
3. A measured value of process-aware physics: how much CALPHAD evaluated along the DED thermal history improves predictions over equilibrium calculations and over data alone.
4. Corrosion data for copper alloys deposited by DED, and a measured link between a one-hour electrochemical screen and immersion tests that take weeks.
5. New copper alloy compositions designed for blown-powder DED, each with its powder specification, deposition settings, tensile properties, corrosion behavior, and repair coupons, using less nickel where the trade-offs allow.

Results that reject a hypothesis are also new knowledge. For example, if H1 fails, the data will show the step whose cost or noise erased the advantage. If H3 fails, they will show which process effects the calculations missed. If H5 fails, they will show which microstructural features the screen missed.

### 1.10 Future Naval Relevance

**ONR S&T Focus Areas.** The primary focus area is Manufacturing, research area Integrated Manufacturing and Sustainment Technologies. The secondary focus area is Materials and Electronics, research area Sustainable and Operationally Robust Materials. Within Code 332 the project fits the program Materials & Processes for Additive Manufacturing [121]. That program lists five thrusts: "(i) rapid qualification frameworks; (ii) tailored AM materials and processes; (iii) large scale AM; (iv) additive repair and (v) AM and digital logistics" [121]. Its research challenges include to "develop and validate computational approaches for AM materials and processes for engineering applications" and to "develop large-scale AM processes for naval materials of interest" [121]. The project serves thrusts (ii) to (iv) with an alloy designed for a large-scale process that also repairs, and thrust (i) with a search method that decides which test to run next and states its confidence in the outcome. It is ICME for additive manufacturing, the approach that the ONR Structural Metals program calls "an efficient rigorous framework for optimizing processing and manufacturing schemes" [122]: computation first, and experiments where they change a decision. It also fits two other Code 332 programs. Corrosion Science and Corrosion Control Technologies asks for a "fundamental understanding of corrosion mechanism in additive manufactured (AM) alloys" [123]. The thrusts of Computer-Aided Materials Design include high-throughput screening concepts and informatics [124].

**Copper alloys, castings, and repair.** The slow production of large castings "has historically gated progress and throughput for both the Columbia and Virginia submarine programs" [1], and the Navy is replacing castings with printed parts. In August 2026 it set out "a streamlined, repeatable technical pathway to substitute legacy cast and wrought parts with 3D-printed metallic materials" [125]. A printed copper-nickel deck drain, made "to provide an alternative to traditional copper-nickel castings," has been installed on a new-construction submarine [126, 127], and the Navy Maritime Industrial Base program has made copper-nickel ship-service ball valves by wire-arc deposition [128]. These parts use the cast compositions, as does ONR-funded research on wire-arc 70-30 copper-nickel [38]. A copper alloy designed for blown-powder DED, one that resists seawater as deposited, would let a shipyard or depot build a part that now waits on a casting. It would also let them restore a worn part in place without the heat treatment the cast alloys need. The ONR corrosion program asks for this kind of work: "modify the chemistry of feed stock materials to enhance AM processing and corrosion resistance of additive-manufactured alloys" [123].

**The longer horizon.** The project designs for single-stream, pre-alloyed powder, which existing DED systems can use now. The same models can later choose blends of elemental and master-alloy powders fed from several hoppers and alloyed in the melt pool, which would let a depot make a new composition without a new powder lot. The guide of Task 4 will describe that extension. The method needs only a network of steps, their costs, and their measurements, so it also applies to wire-arc and laser-wire deposition and to other Navy alloys, including titanium. A plasma melting module for our atomizer (not part of this request) would extend the powder work to alloys that melt above the limit of the induction module.

**Navy engagement and people.** The Naval Surface Warfare Center (NSWC) Carderock Division is the Navy lead for additive manufacturing technology and the NAVSEA lead for welding process and material specifications [129], and it runs marine corrosion laboratories [112]. Its engineers have studied how processing sets the microstructure of NAB [24] and published the temperature-

dependent properties of copper-nickel and wire-arc NAB that our heat flow model uses [69, 70, 72]. The PI presented autonomous experimentation for alloy discovery at Carderock on 14 October 2026, hosted by Dr. Nicholas J. Jones, to an audience of 54 [TODO Sterling: name any Carderock contacts who agree to be listed, and confirm Dr. Jones's title and code]. Each year we will send the candidates and data to materials and corrosion engineers at Carderock and ask them which flowing-seawater and repair tests the final candidates should get. We will visit in Years 2 and 3 (both trips are in the budget). The PI will apply to the Summer Faculty Research Program to spend a summer at Carderock [TODO Sterling: confirm; the 2027 application closes 14 December 2026; Carderock's SFRP topics include sustainment, repair, and corrosion science]. The project will train two Ph.D. students and undergraduates in powder making, deposition, characterization, corrosion testing, and autonomous experimentation. Undergraduates join from our research group and from the courses the PI teaches [TODO Sterling: name the courses]. We will encourage them to apply to the Naval Research Enterprise Internship Program and the SMART and NDSEG programs, which the NOFO names as goals of the YIP.

### 1.11 Project Schedule and Milestones

Table 2 lists the schedule by year. The award start date is 1 April 2027.

### 1.12 Goals and Benchmarks

Table 3 lists each goal and the benchmark used to measure progress. Progress will be reported at each ONR program review.

### 1.13 Gold Standard Science

Executive Order 14303 lists nine tenets of Gold Standard Science [130]. Such science is reproducible, transparent, communicative of error and uncertainty, collaborative and interdisciplinary, and skeptical of its findings and assumptions. It is also structured for falsifiability of hypotheses, subject to unbiased peer review, accepting of negative results as positive outcomes, and without conflicts of interest. The plan follows each point. Each hypothesis above has a stated rejection criterion, and we fix the targets and baselines before each test. Every prediction carries an uncertainty, and the calibration tests (H2, H3) check whether that uncertainty is honest. The random audit of Advance 5 tests the assumptions behind each screen. Replicate deposits and reference specimens measure scatter so that it is not mistaken for a real effect. We release code, data, and models each year so others can rerun every analysis. We will publish results that reject a hypothesis like any other result. The record of the PI supports this: the PI co-authored a review of self-driving laboratories and led open benchmark and software projects (see Section 3).

## 2. Management Approach

This is a single-PI project with no co-PIs or subawards. The PI will lead the method development, set the campaign plans, and supervise all students. Ph.D. student 1 will lead Task 1 (methods, physics inputs, benchmarks, software). Ph.D. student 2 will lead Task 2 (powder, deposits, microscopy, tensile and corrosion tests). An undergraduate researcher will help make and check powder, prepare specimens, and run the corrosion screen. Both Ph.D. students share Task 3, so the methods are tested on the data they will face.

**Table 2.** Schedule and milestones (Y = project year, starting 1 April 2027).

| Year | Work | Milestone (month) |
|---|---|---|
| Y1 | Derive the rule for several objectives; build simulated benchmarks and benchmarks from past data; build the process-aware physics inputs; first copper powder lots in the DED size range and their tracks at Utah; deposit the commercial baselines at both facilities and replicate them to measure scatter; set up the corrosion screen on wrought and cast references; automate image analysis. | M1: first copper powder lot sieved, checked, and melted in tracks at Utah (4).<br>M2: baseline deposits and scatter measured (7).<br>M3: corrosion screen repeatability measured (8).<br>M4: rule for several objectives passes small-problem checks (9).<br>M5: benchmarks frozen and posted (12). |
| Y2 | Derive the rule for pending results; first live campaign with tensile, corrosion, and repair branches; first 30-day immersions; test H2 and H3 on unseen deposits. | M6: first live campaign done (18).<br>M7: H2 and H3 tested (22).<br>M8: methods paper and software release 1 (24). |
| Y3 | Matched live campaigns against the full-chain baseline; test H1, H4, and H5; release software, data, and guide; share powder, deposition, corrosion, and repair data with Navy engineers. | M9: matched campaigns done (32).<br>M10: H1, H4, and H5 tested (34).<br>M11: final release and report (36). |

We buy DED time from the University of Utah and the Colorado School of Mines, and any outside tests from a commercial laboratory. None is a subaward. A BYU student runs the Utah sessions after training by the Utah group, and staff at Mines run each session there. Both follow the settings the search chooses. [TODO Sterling: if Fang or Sam agrees to be named as an unfunded collaborator, say so here and list them under current and pending support (NOFO p. 38)]. Our budget buys about eight half-day sessions a year at Utah and 11, 13, and 9 sessions at Mines in Years 1 to 3. [TODO Sterling: the Utah line is an allowance until Fang sets a rate; dropping the $18,000 potentiostat (Field 11) would buy about 14 more Mines sessions in Year 1]. All other work is at BYU: our atomizer laboratory, the BYU Microscopy Center (SEM, EDS, EBSD), and BYU high-performance computing. The team meets weekly to review new results and choose the next experiments. A shared database records every step, so the search always sees the latest results. The schedule and milestones are in Table 2.

**Risks and responses.** If the rule for several objectives is too slow to compute, we will use a cheaper approximation and measure the loss on the simulated benchmark. If deposit scatter is large, we will add replicates and enter the scatter in the models directly. If a composition cannot be atomized, for example because it melts too close to the limit of the module, the search records the failure as a result, since predicting which compositions can be made is part of the task. If the coating of a

**Table 3.** Goals and the benchmarks that measure them.

| Goal | Benchmark |
|---|---|
| Cost-aware rule for several objectives | Matches exhaustive calculations on small problems; finds a larger set of trade-offs per dollar than a full-chain search on the simulated benchmark. |
| Rule for pending results | Same final quality as the one-at-a-time rule in less calendar time on the simulated benchmark with measured delays (H4). |
| Calibrated predictions | Observed success rates within 10 percentage points of predicted probabilities on unseen deposits (H2). |
| Process-aware physics | Lower calibration error with thermal-history inputs than with equilibrium inputs, and with equilibrium inputs than without (H3). |
| Powder for blown-powder DED | Every deposited composition made in-house, sieved to the DED size range, with its size distribution, flow, and oxygen and carbon content recorded. |
| Corrosion screen | Top-ranked candidates pass the 30-day immersion test more often than bottom-ranked ones (H5). |
| Lower cost on real deposits | Lower median cost to reach the target than the full-chain baseline in matched live campaigns (H1). |
| New alloys | At least [TODO Sterling: number] new compositions deposited dense and crack-free, with tensile, corrosion, and repair-coupon data, beyond the commercial baselines. |
| Open release | Software, data, and models posted each year with tests and documentation. |

plate wears and the melt stops wetting it, we replace the plate. The tungsten-based coating, the best balanced in a comparison by the manufacturer [98], is the alternative. If melts that contain nickel or iron still pick up carbon, which the carbon analysis of each lot will show, we will recoat the crucible before every run instead of every few. We will also replace the graphite rod and the carbon-carbon plate with an alumina sealing rod and a molybdenum-alloy plate [131]. If a lot flows poorly, the search sees that at the powder step and can change the atomizer settings before paying for a deposit. The pre-placed tracks on cast NAB and copper-nickel plate show early whether the first layer fuses to thick plate. If it does not, we will preheat the repair coupons or make them thinner. If 500 W at Utah cannot melt a composition well, its tracks move to Mines at higher cost, and the search learns which compositions need the larger system. If making 2 kg lots slows the campaigns, the Utah step will carry more of the screening, and fewer compositions will be scaled up. If either facility becomes unavailable, the other takes its work at its own cost, which the search already knows. If DED turnaround is slow, the rule for pending results is designed for exactly that case. If the electrochemical screen predicts the immersion tests poorly (H5 rejected), the search will rely more on immersion, at higher cost, and the audited candidates will show which features the screen missed.

## 3. Principal Investigator Qualifications

Sterling G. Baird received a Ph.D. in Materials Science and Engineering from the University of Utah in August 2023. He has been an Assistant Professor of Mechanical Engineering at BYU since December 2025, his first tenure-track position. Before BYU, he was Director of Training and Programs and then a Principal Investigator and Staff Scientist at the Acceleration Consortium, University of Toronto, where he built the development laboratory of the Acceleration Consortium, including about $600,000 of equipment.

The PI combines Bayesian optimization with laboratory hardware. He developed Honegumi, which helps experimentalists set up Bayesian optimization, consulted for Meta on the version 1.0 release of the Ax platform, and organized an international Bayesian optimization hackathon. He co-authored the structural materials section of the *Chemical Reviews* survey of self-driving laboratories and led a review of low-cost self-driving laboratories. His graduate work used Bayesian inference with propagated uncertainty to model grain boundaries. He also designed a semi-automated induction furnace for temperatures up to 2500 °C. At BYU he supervises eight undergraduate researchers and has set up the CALPHAD workflow, the atomizer, the powder doser, and the powder-to-part demonstration described in Section 1.5. [TODO Sterling: add any corrosion or electrochemistry experience; if there is none, say how the students will learn the corrosion methods, e.g., a BYU colleague's training or a Navy laboratory visit. Add any DED or welding experience.] The CV lists all publications and presentations.

**Principal Investigator's Curriculum Vitae**

[TODO Sterling: insert the full CV PDF here.]

**Letter of Support**

[TODO Sterling: insert the signed letter from the BYU official here as a PDF page (draft in support-letter-draft.md; it names aluminum, so revise it for this version).]

## 4. Responsibility

**Resources.** Our laboratory at BYU holds the ultrasonic atomizer, the powder doser, the liquid-handling gantry, and the equipment for powder handling and specimen preparation [TODO Sterling: list the main items bought with start-up funds]. A potentiostat for the corrosion screen is requested in Year 1. The BYU Microscopy Center provides SEM, EDS, and EBSD, and BYU provides high-performance computing. We buy DED sessions at Utah and Mines (Task 2). Tensile tests use the Instron load frames in the materials testing laboratory of the Department of Mechanical Engineering [TODO Sterling: confirm].

**Ability to comply.** The current commitments of the PI are teaching, start-up research, and the pending requests listed under current and pending support. The effort proposed here fits within those commitments.

**Performance history.** [TODO Sterling: list any federal awards managed as PI or key person; if none, state that this would be the PI's first federal award as PI and that BYU Research Administration manages the award.]

**Integrity and business ethics.** The PI has no findings of research misconduct and follows BYU policies on research integrity and conflicts of interest. [TODO Sterling: confirm.]

**Qualifications and eligibility.** The PI received his Ph.D. after 1 January 2019, holds his first full-time tenure-track position, and has not received an ONR YIP award. [TODO Sterling: confirm U.S. citizenship or permanent residency.] BYU is registered in SAM.gov (UEI JWSYC7RUMJD1, CAGE 52534) [TODO Sterling: BYU's SAM registration is listed as ending 4 March 2027, before the award start; confirm that RAO will renew it].

**Organization and controls.** BYU Research Administration manages federal awards, with accounting, purchasing, and property control systems [TODO Sterling: confirm wording with BYU Research Administration]. Laboratory safety follows BYU Risk Management and Safety procedures. We handle metal powders under the written powder safety plan of the lab. Nickel-bearing powders are handled in a fume hood or glove bag with respiratory protection [TODO Sterling: confirm the plan and the handling steps for nickel powder]. Corrosion tests use salt solutions. The dilute acids used to clean immersion coupons [52] are handled in a fume hood under BYU chemical safety procedures.

## 5. Data Management Plan

**Types of data, software, and other materials.** We will produce thermodynamic and heat flow calculation results, and atomizer settings with records of powder size, flow, density, oxygen, and carbon. Settings and session logs cover the pre-placed tracks and DED deposits. Microscopy gives optical and electron micrographs, EDS spectra, and EBSD maps. Mechanical tests give tensile and hardness data, and corrosion tests give polarization and impedance data, immersion mass losses, and cross-sections of corrosion attack. The software products are the search software, benchmark problems, and trained models.

**How the data will be acquired.** Calculations run on BYU high-performance computing. We make powder in our laboratory. DED sessions are bought as a service (Section 2). Microscopy, corrosion screens, immersion tests, and mechanical tests are done at BYU.

**Time and location.** Data are collected at BYU and at the two DED facilities (University of Utah and Colorado School of Mines) from April 2027 to March 2030.

**Processing.** Scripts kept under version control turn raw images and test records into derived quantities (porosity, crack length, dilution, grain size, phase fractions, strength, corrosion rate, attack depth) with stated uncertainty. Each derived value links to its raw files.

**File formats and naming.** Open formats: CSV and JSON for tables, TIFF for images, and the instruments' native formats alongside open exports. Each sample receives an identifier that encodes powder lot, deposition session, and specimen, used in every file name.

**Quality assurance and control.** Replicate deposits measure scatter. We measure reference specimens of wrought 70-30 copper-nickel and cast NAB in every round of corrosion screening. Scripts are tested, and the same benchmarks are rerun before each software release.

**Existing data.** The benchmarks built from past data use our own aluminum data from 2026, with its origin recorded in the metadata.

**Standards.** Metadata will record composition, powder properties, process settings, instrument settings, and the ASTM method behind each test in a documented schema [TODO Sterling: choose a schema, e.g., one used by the NIST AM Bench or Materials Data Facility].

**Preservation.** We will deposit data and software in public repositories (for example, Zenodo with a DOI, and GitHub for code) at each annual release and keep them for at least 10 years. The raw files of each finalist will also go to Materials Commons, as for the Carderock HSLA-80 property database [132].

**Balance of value and cost.** We will keep raw session logs and micrographs in full. Very large intermediate files that scripts can regenerate will not be archived.

**Release restrictions.** Results are expected to be fundamental research. Any data that cannot be cleared for public release under DoD Instruction 5230.09 will not be made public.

**Current and Pending Support (summary)**

**Current.** [TODO Sterling: list current awards, if any, including BYU internal funds that must be reported under the common form.]

**Pending.** AFOSR FY2027 Young Investigator Program, white paper submitted 9 October 2026 (NOFO FA955026S0003, Dynamic Data and Information Processing, Dr. Erik Blasch). Title: "Dynamic Data-Driven Selection of Experiments for Laser-Printed Aluminum Alloys." Up to $150,000 per year for 36 months. The two share the search method. This proposal applies it to copper alloys for blown-powder DED, with corrosion and repair branches. The AFOSR work applies it to aluminum alloys for laser powder bed fusion, so the experimental work does not overlap. [TODO Sterling: state how the method work (Task 1) would be divided if both are selected. Add any other proposals submitted before 30 October, and list collaborators, even unfunded ones (NOFO p. 38).]

## References


[1] NAVSUP Weapon Systems Support, "Shattering bottlenecks: NAVSUP WSS partners with DLA and NAVSEA on submarine advanced manufacturing," DVIDS (30 Jun 2026). https://www.dvidshub.net/news/569018/shattering-bottlenecks-navsup-wss-partners-with-dla-and-navsea-submarine-advanced-manufacturing

[2] MIL-T-16420K, *Tube, Copper-Nickel Alloy, Seamless and Welded (Copper Alloy Numbers 715 and 706)*, Naval Sea Systems Command (validation notice 18 Feb 2026). https://quicksearch.dla.mil/qsDocDetails.aspx?ident_number=11860

[3] MIL-B-24480A, *Bronze, Nickel-Aluminum (UNS No. C95800) Castings for Seawater Service*, Naval Sea Systems Command (20 Jun 1985; validation notice 24 Dec 2020). https://quicksearch.dla.mil/qsDocDetails.aspx?ident_number=17349

[4] Copper Development Association, *Guide to Nickel Aluminium Bronze for Engineers*, CDA Publication 222 (January 2016). https://www.copper.org/applications/marine/nickel_al_bronze/pub-222-nickel-al-bronze-guide-engineers.pdf

[5] A. Saboori, A. Aversa, G. Marchese, S. Biamino, M. Lombardi, and P. Fino, "Application of directed energy deposition-based additive manufacturing in repair," *Appl. Sci.* **9**(16), 3316 (2019). doi:10.3390/app9163316

[6] X. Cao, P. Wanjara, J. Gholipour, and Y. Wang, "Laser-additive repair of cast Ni-Al-bronze components," in *TMS 2019 148th Annual Meeting & Exhibition Supplemental Proceedings* (Springer, 2019), pp. 205-216. doi:10.1007/978-3-030-05861-6_19

[7] Y. Liu, Z. Ye, X. Wang, B. Liang, and Y. Zhang, "Microstructure and mechanical behavior of Cu-9Al-4Ni-3.5Fe-0.5Mn alloy fabricated by laser melting deposition," *Mater. Sci. Eng. A* **826**, 142006 (2021). doi:10.1016/j.msea.2021.142006

[8] T. Dobson, N. Larrosa, M. Reid, K. Rajamudili, S. Ganguly, and H. Coules, "Corrosion mechanisms of plasma welded nickel aluminium bronze immersed in seawater," *Corros. Sci.* **232**, 112004 (2024). doi:10.1016/j.corsci.2024.112004

[9] MIL-PRF-32816, *Nickel Aluminum Bronze, Wire Arc, Directed Energy Deposition, Additively Manufactured*, Naval Sea Systems Command (29 Jun 2026). https://quicksearch.dla.mil/qsDocDetails.aspx?ident_number=287090

[10] MIL-PRF-32804, *Copper-Nickel (70-30), Wire Arc, Directed Energy Deposition, Additively Manufactured*, Naval Sea Systems Command (27 Oct 2025). https://quicksearch.dla.mil/qsDocDetails.aspx?ident_number=286541

[11] MIL-PRF-32815, performance specification for additively manufactured 70-30 copper-nickel made by laser powder bed fusion, Naval Sea Systems Command (29 Jun 2026). https://quicksearch.dla.mil/qsDocDetails.aspx?ident_number=287089

[12] Z. Jia, J. Yang, R. Li, Z. Wang, C. Tan, G. Sun, and E.-H. Han, "Subtractive composition design of nickel-aluminum bronze for laser additive manufacturing: Achieving superior as-deposited mechanical and corrosion performance," *Addit. Manuf.* **127**, 105291 (2026). doi:10.1016/j.addma.2026.105291

[13] A. L. Ma, S. L. Jiang, Y. G. Zheng, and W. Ke, "Corrosion product film formed on the 90/10 copper-nickel tube in natural seawater: Composition/structure and formation mechanism," *Corros. Sci.* **91**, 245-261 (2015). doi:10.1016/j.corsci.2014.11.028

[14] O. O. Ekerenam, A.-L. Ma, Y.-G. Zheng, S.-Y. He, and P. C. Okafor, "Evolution of the corrosion product film and its effect on the erosion-corrosion behavior of two commercial 90Cu-10Ni tubes in seawater," *Acta Metall. Sin. (Engl. Lett.)* **31**(11), 1148-1170 (2018). doi:10.1007/s40195-018-0745-1

[15] C. Ancalmo and S. P. Narra, "Prediction of the powder catchment efficiency and melt track height in laser directed energy deposition," *J. Manuf. Process.* **137**, 207-220 (2025). doi:10.1016/j.jmapro.2025.01.039

[16] A. Singh, S. Kapil, and M. Das, "A comprehensive review of the methods and mechanisms for powder feedstock handling in directed energy deposition," *Addit. Manuf.* **35**, 101388 (2020). doi:10.1016/j.addma.2020.101388

[17] V. Müller, J. M. Fasselt, T. Kruse, C. Klötzer, R. Kleba-Ehrhardt, T. Choma, M. Biegler, and M. Rethmeier, "Using ultrasonic atomization to recycle aluminium bronze chips for additive laser directed energy deposition," *IOP Conf. Ser.: Mater. Sci. Eng.* **1296**, 012036 (2023). doi:10.1088/1757-899X/1296/1/012036

[18] E. Uhlmann, J. Polte, J. M. Fasselt, V. Müller, C. Klötzer-Freese, R. Kleba-Ehrhardt, M. Biegler, and M. Rethmeier, "A comparative evaluation of powder characteristics of recycled material from bronze grinding chips for additive manufacturing," *Materials* **17**(14), 3396 (2024). doi:10.3390/ma17143396

[19] M. Cabrioli, M. Silva Colmenero, S. Gholamzadeh, M. Vanazzi, S. Amirabdollahian, M. Perini, W. Łacisz, and B. Kalicki, "Upcycling of copper scrap into high-quality powder for additive manufacturing: Processing, characterization, and sustainability assessment," *J. Manuf. Mater. Process.* **9**(9), 320 (2025). doi:10.3390/jmmp9090320

[20] Y. Zhu, X. Chen, P. Liu, S. Fu, H. Zhou, and J. Wu, "Effect of iron on the composition and structure of corrosion product film formed in 70/30 copper-nickel alloy," *Anti-Corros. Methods Mater.* **68**(2), 113-121 (2021). doi:10.1108/ACMM-01-2021-2422

[21] J. A. Wharton, R. C. Barik, G. Kear, R. J. K. Wood, K. R. Stokes, and F. C. Walsh, "The corrosion of nickel-aluminium bronze in seawater," *Corros. Sci.* **47**(12), 3336-3367 (2005). doi:10.1016/j.corsci.2005.05.053

[22] E. A. Culpan and G. Rose, "Corrosion behaviour of cast nickel aluminium bronze in sea water," *Br. Corros. J.* **14**(3), 160-166 (1979). doi:10.1179/bcj.1979.14.3.160

[23] G. Sander, J. Tan, P. Balan, O. Gharbi, D. R. Feenstra, L. Singer, S. Thomas, R. G. Kelly, J. R. Scully, and N. Birbilis, "Corrosion of additively manufactured alloys: A review," *Corrosion* **74**(12), 1318-1350 (2018). doi:10.5006/2926

[24] S. M. Orzolek, J. K. Semple, and C. R. Fisher, "Influence of processing on the microstructure of nickel aluminum bronze (NAB)," *Addit. Manuf.* **56**, 102859 (2022). doi:10.1016/j.addma.2022.102859

[25] T. Murray, S. Thomas, Y. Wu, W. Neil, and C. Hutchinson, "Selective laser melting of nickel aluminium bronze," *Addit. Manuf.* **33**, 101122 (2020). doi:10.1016/j.addma.2020.101122

[26] D. Ding, Z. Pan, S. van Duin, H. Li, and C. Shen, "Fabricating superior NiAl bronze components through wire arc additive manufacturing," *Materials* **9**(8), 652 (2016). doi:10.3390/ma9080652

[27] C. Dharmendra, A. Hadadzadeh, B. S. Amirkhiz, G. D. Janaki Ram, and M. Mohammadi, "Microstructural evolution and mechanical behavior of nickel aluminum bronze Cu-9Al-4Fe-4Ni-1Mn fabricated through wire-arc additive manufacturing," *Addit. Manuf.* **30**, 100872 (2019). doi:10.1016/j.addma.2019.100872

[28] B. Li, H. Zheng, C. Han, and K. Zhou, "Nanotwins-containing microstructure and superior mechanical strength of a Cu-9Al-5Fe-5Ni alloy additively manufactured by laser metal deposition," *Addit. Manuf.* **39**, 101825 (2021). doi:10.1016/j.addma.2020.101825

[29] K. Morshed-Behbahani, A. J. Rayner, D. P. Bishop, and A. Nasiri, "Perspectives on the unusual electrochemical corrosion of nickel aluminum bronze (NAB) alloy fabricated through laser-powder bed fusion additive manufacturing," *Corros. Sci.* **228**, 111846 (2024). doi:10.1016/j.corsci.2024.111846

[30] X. Cai, M. Yang, S. Wang, Z. Wang, J. Zhou, and F. Xue, "Experimental investigations on corrosion behavior and antibacterial property of nickel-aluminum bronze fabricated through wire-arc additive manufacturing (WAAM)," *Corros. Sci.* **214**, 111040 (2023). doi:10.1016/j.corsci.2023.111040

[31] A. Fayanas, A. Veiga, A. Perez, A. Concustell, S. Molas, and N. Ordas, "Nickel aluminium bronze alloys with enhanced erosion-corrosion resistance and mechanical strength," *J. Jpn. Soc. Powder Powder Metall.* **72**(Suppl.), S819-S826 (2025). doi:10.2497/jjspm.15F-T13-08

[32] A. Shahriari, D. Chalasani, B. Shalchi Amirkhiz, and M. Mohammadi, "Corrosion performance of wire arc additively manufactured NAB alloy," *npj Mater. Degrad.* **7**, 85 (2023). doi:10.1038/s41529-023-00405-x

[33] L. Zhao, K. Li, J. Yao, Y. Yuan, and B. Du, "The laser deposited nickel-aluminum bronze coatings on SUS630 stainless steel and its corrosion resistance in 3.5 wt.% NaCl solution," *Metals* **12**(5), 781 (2022). doi:10.3390/met12050781

[34] C. V. Hyatt, K. H. Magee, and T. Betancourt, "The effect of heat input on the microstructure and properties of nickel aluminum bronze laser clad with a consumable of composition Cu-9.0Al-4.6Ni-3.9Fe-1.2Mn," *Metall. Mater. Trans. A* **29**(6), 1677-1690 (1998). doi:10.1007/s11661-998-0090-5

[35] C.-L. Yao, H.-S. Kang, K.-Y. Lee, J.-G. Zhai, and D.-S. Shim, "A study on mechanical properties of CuNi2SiCr layered on nickel-aluminum bronze via directed energy deposition," *J. Mater. Res. Technol.* **18**, 5337-5361 (2022). doi:10.1016/j.jmrt.2022.04.159

[36] C.-L. Yao and D.-S. Shim, "Corrosive properties of CuNi2SiCr fabricated through directed energy deposition on a nickel-aluminum bronze substrate," *J. Alloys Compd.* **918**, 165776 (2022). doi:10.1016/j.jallcom.2022.165776

[37] C. Yao, S. He, K.-Y. Lee, K.-Y. Shin, and D.-S. Shim, "Microstructural and interfacial characteristics in repair of nickel-aluminum bronze by in-situ synthesis of Cu-Al alloys via directed energy deposition," *Addit. Manuf.* **105**, 104787 (2025). doi:10.1016/j.addma.2025.104787

[38] J. Song, X. A. Jimenez, A. C. To, and Y. Fu, "Microstructures and corrosion properties of wire arc additive manufactured copper-nickel alloys," *Materials* **17**(4), 876 (2024). doi:10.3390/ma17040876

[39] Z. Liu, Q. Zhang, B. Zhang, Y. Wang, P. Zuo, Z. Zhang, J. Wu, X. Yan, Y. Huang, and X. Li, "Investigation on mechanical properties and corrosion behavior of laser powder bed fusion 70/30 copper-nickel alloy," *Corros. Sci.* **232**, 112040 (2024). doi:10.1016/j.corsci.2024.112040

[40] M. Nadimi, J. Song, L. Cheng, and Y. Fu, "Corrosion evaluation and microstructural characteristics of 70/30 copper-nickel alloy fabricated by laser powder bed fusion," *J. Alloys Metall. Syst.* **9**, 100155 (2025). doi:10.1016/j.jalmes.2025.100155

[41] B. Adak, P. Nash, D. Chen, and A. Swiglo, "Microstructural characterization of laser cladding of Cu-30Ni," *J. Mater. Sci.* **40**(8), 2051-2054 (2005). doi:10.1007/s10853-005-1231-5

[42] S. Karnati, F. F. Liou, and J. W. Newkirk, "Characterization of copper-nickel alloys fabricated using laser metal deposition and blended powder feedstocks," *Int. J. Adv. Manuf. Technol.* **103**, 239-250 (2019). doi:10.1007/s00170-019-03553-0

[43] S. Yadav, A. N. Jinoop, N. Sinha, C. P. Paul, and K. S. Bindra, "Parametric investigation and characterization of laser directed energy deposited copper-nickel graded layers," *Int. J. Adv. Manuf. Technol.* **108**, 3779-3791 (2020). doi:10.1007/s00170-020-05644-9

[44] N. Williams, K. Snyder, I. Smith, A. Duong, E. Carpenter, and R. Barua, "The first layer: Single-track insights into direct energy deposition processed Cu-Ni thermoelectric alloys," *J. Manuf. Mater. Process.* **9**(6), 170 (2025). doi:10.3390/jmmp9060170

[45] S. Bhattacharya, G. P. Dinda, A. K. Dasgupta, H. Natu, B. Dutta, and J. Mazumder, "Microstructural evolution and mechanical, and corrosion property evaluation of Cu-30Ni alloy formed by direct metal deposition process," *J. Alloys Compd.* **509**(22), 6364-6373 (2011). doi:10.1016/j.jallcom.2011.03.091

[46] ISSC Committee V.3, "Materials and fabrication technology," in *Proceedings of the 18th International Ship and Offshore Structures Congress*, Vol. 2 (Rostock, 2012), Sec. 3.3.2.

[47] M. Eckstein, "Shipbuilding industry looks to 3D printing to accelerate pace," *Defense News* (2 Feb 2023). https://www.defensenews.com/naval/2023/02/02/shipbuilding-industry-looks-to-3d-printing-to-accelerate-pace/

[48] ASTM D1141, *Standard Practice for the Preparation of Substitute Ocean Water*, ASTM International.

[49] ASTM G59, *Standard Test Method for Conducting Potentiodynamic Polarization Resistance Measurements*, ASTM International.

[50] ASTM G102, *Standard Practice for Calculation of Corrosion Rates and Related Information from Electrochemical Measurements*, ASTM International.

[51] ASTM G31, *Standard Guide for Laboratory Immersion Corrosion Testing of Metals*, ASTM International.

[52] ASTM G1, *Standard Practice for Preparing, Cleaning, and Evaluating Corrosion Test Specimens*, ASTM International.

[53] S. J. Yuan and S. O. Pehkonen, "Surface characterization and corrosion behavior of 70/30 Cu-Ni alloy in pristine and sulfide-containing simulated seawater," *Corros. Sci.* **49**(3), 1276-1304 (2007). doi:10.1016/j.corsci.2006.07.003

[54] T. Dobson, A. Yunnie, D. Kaloudis, N. Larrosa, and H. Coules, "Biofouling and corrosion rate of welded nickel aluminium bronze in natural and simulated seawater," *Biofouling* **40**(2), 193-208 (2024). doi:10.1080/08927014.2024.2326067

[55] M. Iannuzzi, K. L. Vasanth, and G. S. Frankel, "Unusual correlation between SKPFM and corrosion of nickel aluminum bronzes," *J. Electrochem. Soc.* **164**(9), C488-C497 (2017). doi:10.1149/2.0391709jes

[56] U.S. Department of the Interior, "2022 final list of critical minerals," *Federal Register* **87**, 10381 (24 Feb 2022). https://www.federalregister.gov/documents/2022/02/24/2022-04027/2022-final-list-of-critical-minerals

[57] U.S. Department of the Interior, "Final 2025 list of critical minerals," *Federal Register* **90**, 50494 (7 Nov 2025). https://www.federalregister.gov/documents/2025/11/07/2025-19813/final-2025-list-of-critical-minerals

[58] Copper Development Association, alloy data sheets for C70600, C71500, and C95800, physical properties (liquidus and solidus given in °F; read from Internet Archive snapshots of 2025 and 2026). https://alloys.copper.org/alloy/C71500

[59] R. Astudillo and P. I. Frazier, "Bayesian optimization of function networks," *Advances in Neural Information Processing Systems* **34**, 14463-14475 (2021). arXiv:2112.15311

[60] P. Buathong, J. Wan, R. Astudillo, S. Daulton, M. Balandat, and P. I. Frazier, "Bayesian optimization of function networks with partial evaluations," *Proc. 41st Int. Conf. Machine Learning (ICML)*, PMLR **235**, 4752-4784 (2024). proceedings.mlr.press/v235/buathong24a.html

[61] G. B. Olson, "Computational design of hierarchically structured materials," *Science* **277**(5330), 1237-1242 (1997). doi:10.1126/science.277.5330.1237

[62] National Research Council, *Integrated Computational Materials Engineering: A Transformational Discipline for Improved Competitiveness and National Security* (National Academies Press, Washington, DC, 2008). doi:10.17226/12199

[63] J.-O. Andersson, T. Helander, L. Höglund, P. Shi, and B. Sundman, "Thermo-Calc & DICTRA, computational tools for materials science," *Calphad* **26**(2), 273-312 (2002). doi:10.1016/S0364-5916(02)00037-8

[64] R. Otis and Z.-K. Liu, "pycalphad: CALPHAD-based computational thermodynamics in Python," *J. Open Res. Softw.* **5**(1), 1 (2017). doi:10.5334/jors.140

[65] S. Kou, "A criterion for cracking during solidification," *Acta Mater.* **88**, 366-374 (2015). doi:10.1016/j.actamat.2015.01.034

[66] A. M. Jordan and Q. Zhang, "Evaluating thermodynamics-based hot-cracking models for various AM alloys," *Integr. Mater. Manuf. Innov.* **15**, 74-88 (2026). doi:10.1007/s40192-025-00437-y

[67] D. Rosenthal, "The theory of moving sources of heat and its application to metal treatments," *Trans. ASME* **68**(8), 849-865 (1946). doi:10.1115/1.4018624

[68] A. J. Pinkerton, "Advances in the modeling of laser direct metal deposition," *J. Laser Appl.* **27**(S1), S15001 (2015). doi:10.2351/1.4815992

[69] J. E. Norkett and C. R. Fisher, "Temperature-dependent material property database of cast 70/30 Cu-Ni material," *Integr. Mater. Manuf. Innov.* **14**(3), 372-383 (2025). doi:10.1007/s40192-025-00409-2

[70] J. E. Norkett and C. R. Fisher, "Temperature-dependent material property database for 90/10 Cu-Ni plate," *Integr. Mater. Manuf. Innov.* **15**(1), 169-180 (2026). doi:10.1007/s40192-026-00444-7

[71] S. M. Orzolek, J. E. Norkett, and C. R. Fisher, "Temperature-dependent material property database for C63200 nickel-aluminum bronze (NAB) plate," *Integr. Mater. Manuf. Innov.* **12**(4), 481-492 (2023). doi:10.1007/s40192-023-00325-3

[72] J. E. Norkett, J. H. Canaday, S. M. Orzolek, and C. R. Fisher, "Temperature-dependent material property database for nickel-aluminum bronze (NAB) wire-arc additive components," *Integr. Mater. Manuf. Innov.* **15**(1), 224-241 (2026). doi:10.1007/s40192-026-00454-5

[73] G. W. Hatala, Q. Wang, E. W. Reutzel, C. R. Fisher, and J. K. Semple, "A thermo-mechanical analysis of laser hot wire additive manufacturing of NAB," *Metals* **11**(7), 1023 (2021). doi:10.3390/met11071023

[74] L. Torresi and P. Friederich, "Multi-stage Bayesian optimisation for dynamic decision-making in self-driving labs," *Digital Discovery* **5**, 1900-1912 (2026). doi:10.1039/d5dd00572h

[75] S. M. A. A. Alvi, B. Vela, V. Attari, J. Janssen, D. Perez, D. Allaire, and R. Arróyave, "Deep Gaussian process-based cost-aware batch Bayesian optimization for complex materials design campaigns," *npj Comput. Mater.* **12**, 105 (2026). doi:10.1038/s41524-026-01981-7

[76] A. Talbot, S. S. Dash, J. Li, C. Cheng, M. Ramesh, X. Shang, J. Zhang, C. Cojocaru, J. Hattrick-Simpers, and Y. Zou, "Active learning for the accelerated discovery of complex concentrated NiCoCr alloys in additive manufacturing," *npj Adv. Manuf.* (2026), published online 23 Jun 2026. doi:10.1038/s44334-026-00098-5

[77] P. Nelaturu, J. R. Hattrick-Simpers, M. Moorehead, V. Jambur, I. Szlufarska, A. Couet, and D. J. Thoma, "Multi-principal element alloy discovery using directed energy deposition and machine learning," *Mater. Sci. Eng. A* **891**, 145945 (2024). doi:10.1016/j.msea.2023.145945

[78] N. Menon, S. Mondal, and A. Basak, "Multi-fidelity surrogate-based process mapping with uncertainty quantification in laser directed energy deposition," *Materials* **15**(8), 2902 (2022). doi:10.3390/ma15082902

[79] T. M. Smith, C. A. Kantzos, N. A. Zarkevich, B. J. Harder, M. Heczko, P. R. Gradl, A. C. Thompson, M. J. Mills, T. P. Gabb, and J. W. Lawson, "A 3D printable alloy designed for extreme environments," *Nature* **617**, 513-518 (2023). doi:10.1038/s41586-023-05893-0

[80] S. Thapliyal, M. Komarasamy, S. Shukla, L. Zhou, H. Hyer, S. Park, Y. Sohn, and R. S. Mishra, "An integrated computational materials engineering-anchored closed-loop method for design of aluminum alloys for additive manufacturing," *Materialia* **9**, 100574 (2020). doi:10.1016/j.mtla.2019.100574

[81] S. M. Taheri-Mousavi, M. Xu, F. Hengsbach, C. Houser, Z. Ge, B. Glaser, S. Wei, M. Schaper, J. M. LeBeau, G. B. Olson, and A. J. Hart, "Additively manufacturable high-strength aluminum alloys with coarsening-resistant microstructures achieved via rapid solidification," *Adv. Mater.* **38**(4), e09507 (2026), published online 2 Oct 2025. doi:10.1002/adma.202509507

[82] X. Wang and W. Xiong, "Uncertainty quantification and composition optimization for alloy additive manufacturing through a CALPHAD-based ICME framework," *npj Comput. Mater.* **6**, 188 (2020). doi:10.1038/s41524-020-00454-9

[83] F. Markanday, G. Conduit, B. Conduit, J. Pürstl, K. Christofidou, L. Chechik, G. Baxter, C. Heason, and H. Stone, "Design of a Ni-based superalloy for laser repair applications using probabilistic neural network identification," *Data-Centric Eng.* **3**, e30 (2022). doi:10.1017/dce.2022.31

[84] A. Sparrow, W. Smith, and T. Ansell, "LPBF parameter optimization of mechanical, non-thermal generated C71500 powder," *Metals* **16**(1), 10 (2026). doi:10.3390/met16010010

[85] R. Tamura, T. Osada, K. Minagawa, T. Kohata, M. Hirosawa, K. Tsuda, and K. Kawagishi, "Machine learning-driven optimization in powder manufacturing of Ni-Co based superalloy," *Mater. Des.* **198**, 109290 (2021). doi:10.1016/j.matdes.2020.109290

[86] S. Daulton, M. Balandat, and E. Bakshy, "Parallel Bayesian optimization of multiple noisy objectives with expected hypervolume improvement," *Advances in Neural Information Processing Systems* **34** (2021). arXiv:2105.08195

[87] L. Zhang, X. Cai, B. Song, B. Dong, S. Lin, and F. Khoshnaw, "Effect of low-level Ti microalloying on the microstructure, mechanical properties, and corrosion behaviour of WAAM-processed nickel-aluminium bronze," *Adv. Mater. Join.* **1**, 17 (2026). doi:10.1007/s44500-026-00016-x

[88] B. DeCost, H. Joress, S. Sarker, A. Mehta, and J. Hattrick-Simpers, "Towards automated design of corrosion resistant alloy coatings with an autonomous scanning droplet cell," *JOM* **74**, 2941-2950 (2022). doi:10.1007/s11837-022-05367-0

[89] D. Sur, E. F. Holcombe, W. H. Blades, E. A. Anber, D. L. Foley, B. L. DeCost, J. Liu, J. Hattrick-Simpers, *et al.*, "An experimental high-throughput to high-fidelity study towards discovering Al-Cr containing corrosion-resistant compositionally complex alloys," *High Entropy Alloys Mater.* **1**, 336-353 (2023). doi:10.1007/s44210-023-00020-0

[90] P. C. Collins and T. K. Ales, "Additive manufacturing enabled autonomous laboratory for materials discovery and concurrent design of materials and shape," *MRS Bull.* **51**(6), 746-769 (2026). doi:10.1557/s43577-026-01078-y

[91] E. Povoden-Karadeniz, *MatCalc open thermodynamic database for nickel-base alloys (mc_ni)*, version 2.036, TU Wien (2024), Open Database License 1.0. https://www.matcalc.at/index.php/databases/open-databases

[92] Thermo-Calc Software, *TCS Cu-based Alloys Database (TCCU7): Technical Information*, for Thermo-Calc 2026b (2026): 34 elements, 175 assessed binary and 79 assessed ternary systems. https://thermocalc.com/wp-content/uploads/Documentation/Databases/Thermodynamic/tccu7-technical-info.pdf

[93] Department of Metallurgical Engineering, University of Utah, "Direct Energy Deposition," 3D printing research web page (accessed 24 Oct 2026). https://3dprinting.metallurgy.utah.edu/direct_energy_deposition/index.php

[94] J. Snoek, H. Larochelle, and R. P. Adams, "Practical Bayesian optimization of machine learning algorithms," *Advances in Neural Information Processing Systems* **25** (2012). arXiv:1206.2944

[95] T. DebRoy, H. L. Wei, J. S. Zuback, T. Mukherjee, J. W. Elmer, J. O. Milewski, A. M. Beese, A. Wilson-Heid, A. De, and W. Zhang, "Additive manufacturing of metallic components: Process, structure and properties," *Prog. Mater. Sci.* **92**, 112-224 (2018). doi:10.1016/j.pmatsci.2017.10.001

[96] R. Jones, A. Ang, N. Phan, M. R. Brindza, M. B. Nicholas, C. Timbrell, D. Peng, and R. Chandwani, "A review of the parameters controlling crack growth in AM steels and its implications for limited-life AM and CSAM parts," *Materials* **19**(2), 372 (2026). doi:10.3390/ma19020372

[97] Y. Tan, H. Xie, X. Feng, S. Huang, W. Zhang, L. Peng, W. Wang, and Y. Zhao, "Effect of Al/Ni ratio on the microstructure and properties of nickel-aluminum bronze alloys," *Materials* **17**(6), 1330 (2024). doi:10.3390/ma17061330

[98] T. Choma, M. J. Kruszewski, A. Chądzyńska, B. Kalicki, B. Morończyk, J. Ciftci, Ł. Żrodowski, J. Zdunek, and M. Leonowicz, "Design-oriented comparison of Si–Me (Me = Mo, Ti, Zr, Ta, W) infiltration coatings on C/C sonotrodes for ultrasonic atomization of CuSn8: microstructure, phase constitution, wettability, nanoindentation, and process performance," *Materials* **19**(13), 2803 (2026). doi:10.3390/ma19132803

[99] R. J. Lang, "Ultrasonic atomization of liquids," *J. Acoust. Soc. Am.* **34**(1), 6-8 (1962). doi:10.1121/1.1909020

[100] Z. Pi, K. Du, X. Chen, Z. Zheng, and C. Wang, "Laser cladding remanufacturing of propeller ZCuAl8Mn13Fe3Ni2 alloy," *Sci. Rep.* **15**, 12381 (2025). doi:10.1038/s41598-025-96674-4

[101] ASTM B213, *Standard Test Methods for Flow Rate of Metal Powders Using the Hall Flowmeter Funnel*, ASTM International.

[102] ASTM B212, *Standard Test Method for Apparent Density of Free-Flowing Metal Powders Using the Hall Flowmeter Funnel*, ASTM International.

[103] C. R. Fisher, J. J. Mecholsky Jr., H. B. Henderson, M. S. Kesler, and M. V. Manuel, "A reactive element approach to improve fracture healing in metallic systems," *Front. Mater.* **6**, 210 (2019). doi:10.3389/fmats.2019.00210

[104] InssTek, *MX-Lab* product brochure (2024).

[105] ASTM G61, *Standard Test Method for Conducting Cyclic Potentiodynamic Polarization Measurements for Localized Corrosion Susceptibility of Iron-, Nickel-, or Cobalt-Based Alloys*, ASTM International.

[106] G. S. Frankel, "Pitting corrosion of metals: a review of the critical factors," *J. Electrochem. Soc.* **145**, 2186-2198 (1998). doi:10.1149/1.1838615

[107] D. A. Shifler and D. M. Aylor, "Considerations for the testing of materials and components in seawater," CORROSION 2002, NACE International, paper 02217 (2002). doi:10.5006/c2002-02217

[108] D. A. Shifler and D. G. Enos, "Marine corrosion testing," in *LaQue's Handbook of Marine Corrosion* (Wiley, 2022), pp. 379-420. doi:10.1002/9781119788867.ch15

[109] D. A. Shifler, "Understanding and modeling galvanic corrosion in marine environments," CORROSION 2006, NACE International, paper 06283 (2006). doi:10.5006/c2006-06283

[110] P. T. Brewick, "Simulating pitting corrosion in AM 316L microstructures through phase field methods and computational modeling," *J. Electrochem. Soc.* **169**(1), 011503 (2022). doi:10.1149/1945-7111/ac4935

[111] C. M. Hangarter, A. B. Geltmacher, N. R. Tailleart, P. G. Callahan, D. S. Watring, S. M. Olig, and W. Kinee, "X-ray computed microtomography evaluation of electrochemically accelerated stainless steel crevice corrosion," *ECS Adv.* **4**(1), 011501 (2025). doi:10.1149/2754-2734/adb16a

[112] NSWC Carderock Division, *Partnering with Carderock* (2024, Distribution A). https://www.navsea.navy.mil/Portals/103/Documents/NSWC_Carderock/Partnering%20with%20Carderock%20-%202024%20Dist%20A.pdf

[113] P. Du, T. Chen, G. Xie, Q. Wang, and X. Liu, "The difference in anti-corrosion effect between the addition of La and Ce of a B10 alloy," *J. Mater. Res. Technol.* **36**, 8062-8074 (2025). doi:10.1016/j.jmrt.2025.05.090

[114] ASTM B111/B111M, *Standard Specification for Copper and Copper-Alloy Seamless Condenser Tubes and Ferrule Stock*, ASTM International.

[115] ASTM B369/B369M, *Standard Specification for Copper-Nickel Alloy Castings*, ASTM International.

[116] S. A. Campbell, G. J. W. Radford, C. D. S. Tuck, and B. D. Barker, "Corrosion and galvanic compatibility studies of a high-strength copper-nickel alloy," *Corrosion* **58**, 57-71 (2002). doi:10.5006/1.3277305

[117] AWS A5.7/A5.7M, *Specification for Copper and Copper-Alloy Bare Welding Rods and Electrodes*, American Welding Society.

[118] R. O. Lewis and F. L. LaQue, "Evaluation of copper-nickel alloys for OTEC applications," Argonne National Laboratory report ANL/OTEC-BCM-019 (1981). doi:10.2172/6255887

[119] K. G. Reyes, J. Liu, and C. J. Díaz Vargas, "Decision-making under uncertainty for multi-stage pipelines: simulation studies to benchmark screening strategies," *JOM* **74**(8), 2897-2907 (2022). doi:10.1007/s11837-022-05368-z

[120] K.-H. Huang, C. D. Jamieson, and A. Basak, "Uncertainty-aware reconstruction of full process maps in laser-directed energy deposition using scarce data," *Weld. World* (2026), online 26 Sep 2026. doi:10.1007/s40194-026-02629-2

[121] Office of Naval Research, "Materials & Processes for Additive Manufacturing," Code 33 Manufacturing Focus Area (accessed 17 Oct 2026). https://www.onr.navy.mil/organization/departments/code-33/manufacturing-focus-area/additive-manufacturing

[122] Office of Naval Research, "Structural Metals," Code 33 Materials Focus Area (updated November 2020; accessed 17 Oct 2026). https://www.onr.navy.mil/organization/departments/code-33/materials-focus-area/structural-metals

[123] Office of Naval Research, "Corrosion Science and Corrosion Control Technologies," Code 33 Materials Focus Area (accessed 17 Oct 2026). https://www.onr.navy.mil/organization/departments/code-33/materials-focus-area/corrosion-science-and-corrosion-control

[124] Office of Naval Research, "Computer-Aided Materials Design (CAMD)," Code 33 Materials Focus Area (accessed 10 Oct 2026). https://www.onr.navy.mil/organization/departments/code-33/materials-focus-area/computer-aided-materials-design

[125] U.S. Navy, "Navy removes bureaucratic barriers to accelerate 3D printing and accelerate submarine production," press release (21 Aug 2026), on Project Peculiar Document 802-8436658. https://www.navy.mil/Press-Office/Press-Releases/display-pressreleases/Article/4580193/navy-removes-bureaucratic-barriers-to-accelerate-3d-printing-and-accelerate-sub/

[126] HII, "HII partners to advance additive manufacturing on a Virginia-class submarine," news release (25 Sep 2023). https://www.hii.com/news/hii-additive-manufacturing-virginia-class-submarine-newport-news-shipbuilding-2023

[127] Newport News Shipbuilding, "One small part for the Virginia-class submarine program, one giant leap for additive manufacturing," NNS To Go (17 Sep 2025). https://www.nnstogo.com/articles/one-small-part-for-the-virginia-class-submarine-program-one-giant-leap-for-additive-manufacturing/

[128] Navy Maritime Industrial Base Program, "Navy's MIB program celebrates year of focused effort to strengthen maritime industrial capability and capacity," DVIDS (5 Sep 2025). https://www.dvidshub.net/news/549872/navys-mib-program-celebrates-year-focused-effort-strengthen-maritime-industrial-capability-and-capacity

[129] NSWC Carderock Division, "Structures and Materials," core competency web page (accessed 10 Oct 2026). https://www.navsea.navy.mil/Home/Warfare-Centers/NSWC-Carderock/What-We-Do/Core-Competencies/Structures-and-Materials/

[130] Executive Order 14303, "Restoring Gold Standard Science," *Federal Register* **90**, 22601 (29 May 2025). https://www.federalregister.gov/documents/2025/05/29/2025-09802/restoring-gold-standard-science

[131] J. Ciftcli, T. Choma, B. Morończyk, B. Kalicki, F. Puchalski, and Ł. Żrodowski, "Alloy development from sustainable materials – close-loop of materials using ultrasonic atomization," *J. Jpn. Soc. Powder Powder Metall.* **72**(Suppl.), S729-S735 (2025). doi:10.2497/jjspm.15E-SIS13-03

[132] J. K. Semple, D. H. Bechetti, W. Zhang, and C. R. Fisher, "Temperature-dependent material property databases for marine steels, Part 3: HSLA-80," *Integr. Mater. Manuf. Innov.* **11**(4), 648-674 (2022). doi:10.1007/s40192-022-00288-x. Data: Materials Commons, doi:10.13011/m3-cz82-dw37